USED CAR PRICE PREDICTION



A Machine Learning and Streamlit Dashboard Project

Presented by:
1. joy chandra shill
2. ksh chniglamba shingha

Buy now

PROBLEM STATEMENT

Car prices depend on multiple vehicle characteristics.

Manual price estimation can be inconsistent.

Goal: learn patterns from the supplied dataset and estimate a price from vehicle inputs.

The result is a dataset-based estimate, not a guaranteed market quote.



PROJECT OBJECTIVES

- Inspect and prepare the dataset.
- Explore price patterns using charts and summary statistics.
- Train and compare regression models.
- Evaluate performance on unseen test records.
- Deploy an interactive prediction dashboard with Streamlit



Exploratory Data Analysis:

Price Distribution:

Why choose us?

Why Do We Use EDA?

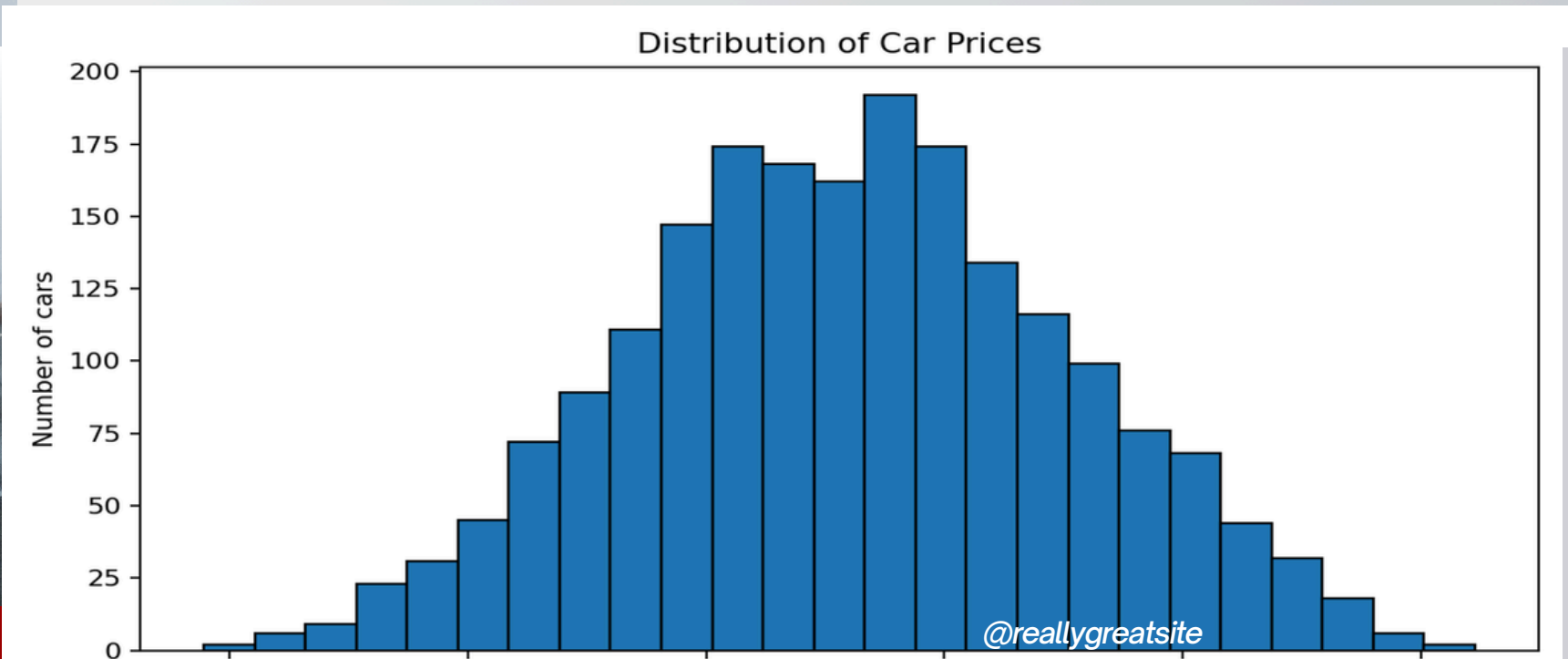
1. To Understand Data: EDA helps us understand the dataset and its features.

2. To Find Missing Values: It helps identify missing or incomplete data.

3. To Detect Outliers: It helps find unusual values in the dataset.

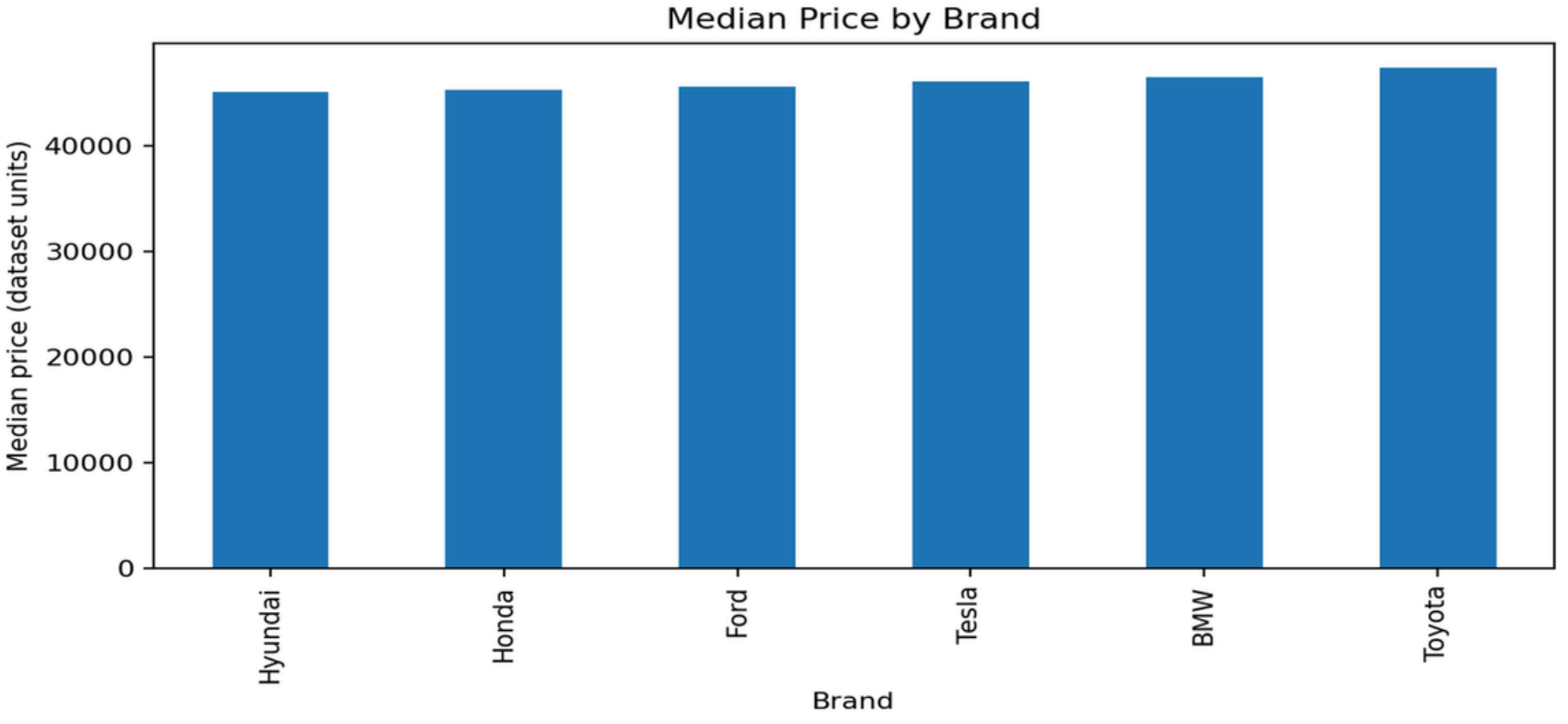
4. To Identify Relationships: It helps us understand relationships between different features.

5. To Improve Model Performance: It helps us prepare data and select suitable features for Machine Learning models.





Exploratory Data Analysis: Median Price by Brand









METHODOLOGY

- Split data into 80% training and 20% testing (random_state = 42).
- Handle missing numeric values with median imputation and categorical values with most-frequent imputation.
- Encode categorical features using one-hot encoding.
- Compare Random Forest, Extra Trees and Gradient Boosting regression.
- Select the model with the lowest MAE on the held-out test set.

jshill668@gmail.com

https://github.com/learnwithjoyshill/

Assam Down Town University

@learn wiith joy shill

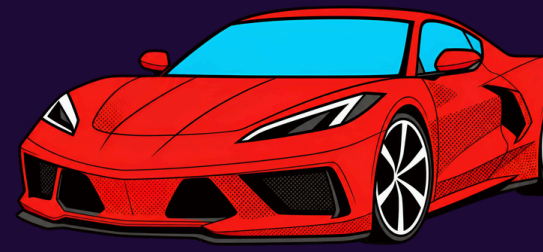
MODEL PERFORMANCE

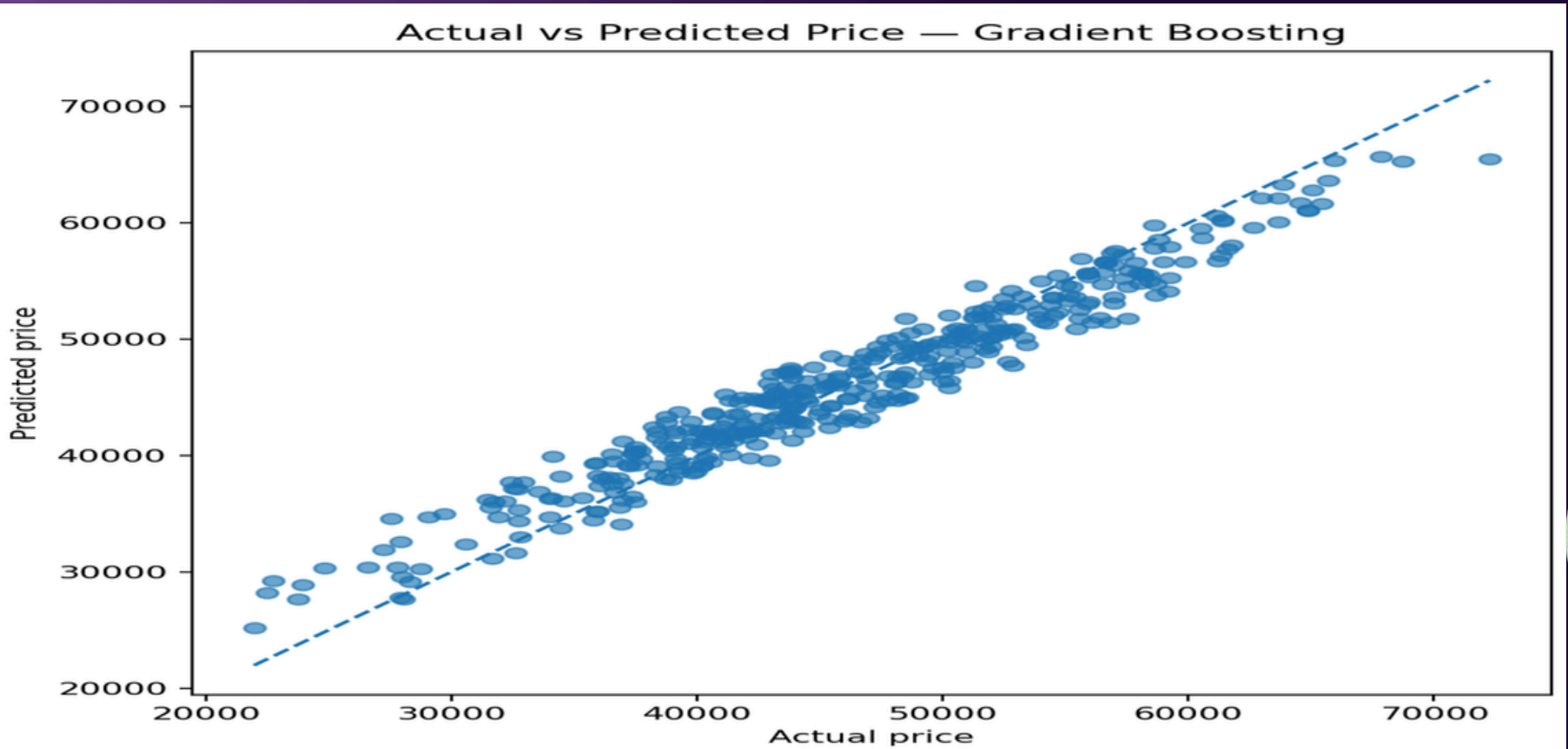


SELECTED MODEL:
GRADIENT BOOSTING
MAE: 1,962.13
RMSE: 2,430.69
R^2: 0.928

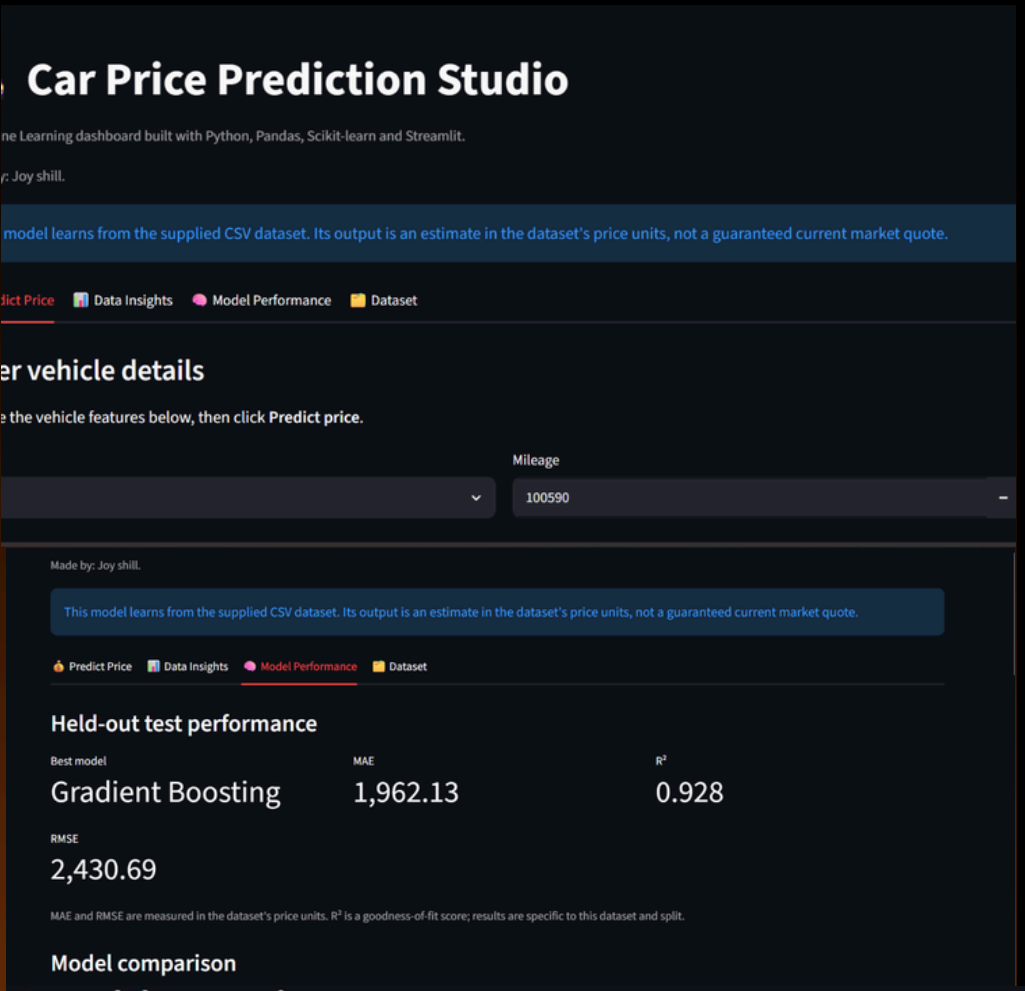
Metrics are based on one held-out test split and use the dataset's price units.

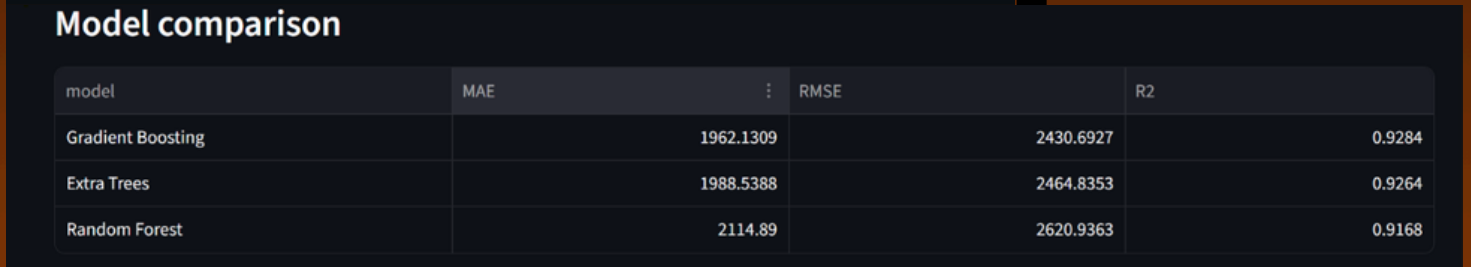
Actual vs Predicted Prices











DASHBOARD FEATURES

- ✓ Enter vehicle specifications in a Streamlit form.
- ✓ Predict a price using the saved preprocessing + model pipeline.
- ✓ View dataset statistics and charts.
- ✓ Compare model metrics and inspect the dataset preview.
- ✓ Download the original CSV from the dashboard.

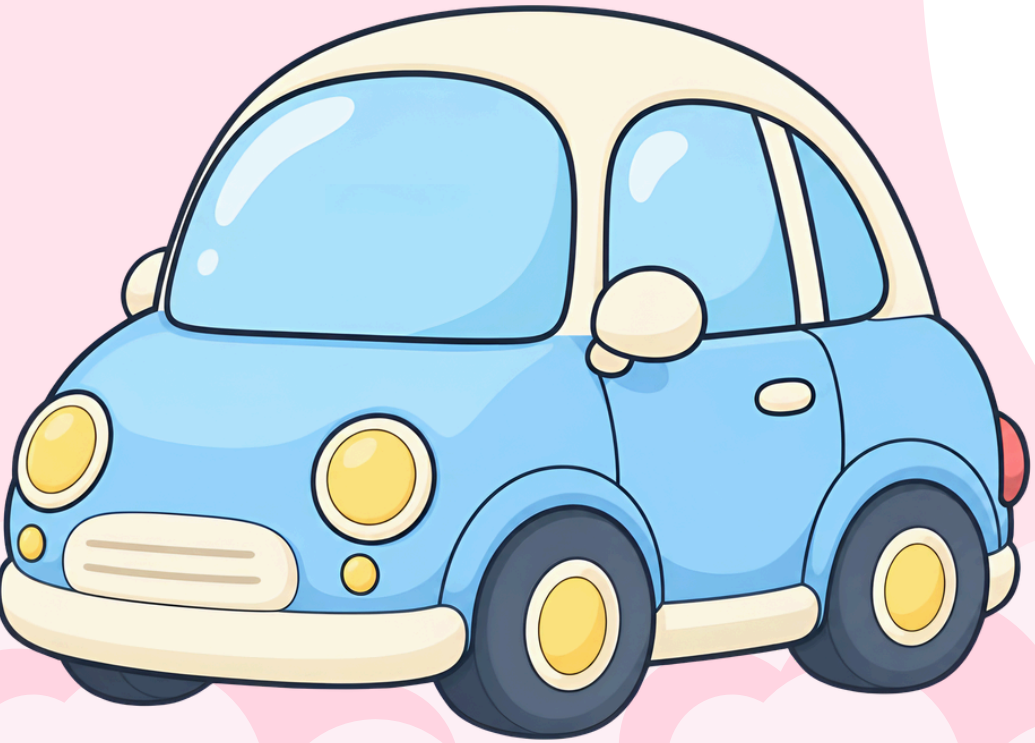
Limitations and Future Work

Dataset source and currency are not specified.
One random split may not reflect performance on other data.
Validate on a larger, real-world dataset before using predictions for decisions.
Future work: cross-validation, hyperparameter tuning, more features, and explainability





Conclusion



Built an end-to-end workflow from CSV data to a trained regression model.
Connected the model to an interactive dashboard.
The project demonstrates data preparation, visualization, model evaluation and deployment.
Thank you — Questions?


Thank
You